

Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-14.jpg, we are tasked with evaluating the following limit:

$$\lim_{x \rightarrow 0} f(x) \quad \text{where} \quad f(x) = \log_{\cos(3x)}(\cos(2ix)) \quad \text{and} \quad i = \sqrt{-1}$$

1. Simplifying the Expression

First, we apply the change of base formula for logarithms to rewrite $f(x)$ using the natural logarithm:

$$f(x) = \frac{\ln(\cos(2ix))}{\ln(\cos(3x))}$$

Next, we use the relationship between the standard cosine function and the hyperbolic cosine function. For any real number u , it is a known identity that $\cos(iu) = \cosh(u)$. Applying this to our numerator with $u = 2x$:

$$\cos(2ix) = \cosh(2x)$$

Substituting this back, the function simplifies to:

$$f(x) = \frac{\ln(\cosh(2x))}{\ln(\cos(3x))}$$

2. Applying Maclaurin Series Expansions

To evaluate the indeterminate limit as $x \rightarrow 0$, we can utilize the Maclaurin series expansions (Taylor series centered at $x = 0$) for the involved functions.

The expansions for the hyperbolic cosine and standard cosine functions are:

$$\begin{aligned} \cosh(u) &= 1 + \frac{u^2}{2} + \mathcal{O}(u^4) \\ \cos(v) &= 1 - \frac{v^2}{2} + \mathcal{O}(v^4) \end{aligned}$$

Substituting $u = 2x$ and $v = 3x$:

$$\begin{aligned} \cosh(2x) &= 1 + \frac{(2x)^2}{2} + \mathcal{O}(x^4) = 1 + 2x^2 + \mathcal{O}(x^4) \\ \cos(3x) &= 1 - \frac{(3x)^2}{2} + \mathcal{O}(x^4) = 1 - \frac{9x^2}{2} + \mathcal{O}(x^4) \end{aligned}$$

Next, we apply the Taylor expansion for the natural logarithm, $\ln(1+z) = z - \frac{z^2}{2} + \mathcal{O}(z^3)$, as $z \rightarrow 0$:

$$\begin{aligned} \ln(\cosh(2x)) &= \ln(1 + 2x^2 + \mathcal{O}(x^4)) = 2x^2 + \mathcal{O}(x^4) \\ \ln(\cos(3x)) &= \ln\left(1 - \frac{9x^2}{2} + \mathcal{O}(x^4)\right) = -\frac{9x^2}{2} + \mathcal{O}(x^4) \end{aligned}$$

3. Evaluating the Limit

We now substitute these series approximations back into our limit expression:

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{2x^2 + \mathcal{O}(x^4)}{-\frac{9x^2}{2} + \mathcal{O}(x^4)}$$

Factoring out x^2 from both the numerator and the denominator yields:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{x^2(2 + \mathcal{O}(x^2))}{x^2(-\frac{9}{2} + \mathcal{O}(x^2))} \\ &= \lim_{x \rightarrow 0} \frac{2 + \mathcal{O}(x^2)}{-\frac{9}{2} + \mathcal{O}(x^2)} \end{aligned}$$

As $x \rightarrow 0$, the remaining $\mathcal{O}(x^2)$ terms approach 0, leaving us with:

$$\lim_{x \rightarrow 0} f(x) = \frac{2}{-\frac{9}{2}} = 2 \cdot \left(-\frac{2}{9}\right) = -\frac{4}{9}$$

Conclusion

The final evaluated limit is:

$$\lim_{x \rightarrow 0} \log_{\cos(3x)}(\cos(2ix)) = -\frac{4}{9}$$